Knowledge-Based Systems 33 (2012) 145–151

Contents lists available at SciVerse ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

Probabilistic outputs for twin support vector machines

Yuan-Hai Shao [a,*], Nai-Yang Deng [b], Zhi-Min Yang [a], Wei-Jie Chen [a], Zhen Wang [c]

[a] Zhijiang College, Zhejiang University of Technology, Hangzhou 310024, PR China
[b] College of Science, China Agricultural University, Beijing 100083, PR China
[c] College of Mathematics, Jilin University, Changchun 130012, PR China

ARTICLE INFO

Article history:
Received 9 November 2011
Received in revised form 13 March 2012
Accepted 1 April 2012
Available online 7 April 2012

Keywords:
Pattern classification
Probabilistic outputs
Support vector machines
Twin support vector machines
Non-parallel hyperplanes

ABSTRACT

In many cases, the output of a classifier should be a calibrated posterior probability to enable post-processing. However, twin support vector machines (TWSVM) do not provide such probabilities. In this paper, we propose a TWSVM probability model, called PTWSVM, to estimate the posterior probability. Note that our model is quite different from the SVM probability model because of the different mechanism of TWSVM and SVM. In fact, for TWSVM, we first define a new ranking continues output by comparing the distances between the sample and the two non-parallel hyperplanes, and then map this ranking continues output into probability by training the parameters of an additional sigmoid function. Our PTWSVM has been tested on both artificial datasets and several data-mining-style datasets, and the numerical experiments indicate that PTWSVM yields nice results.

1. Introduction

Constructing a classifier producing a posterior probability $P(\text{class}|\text{input})$ is very useful in practical recognition problems [1–3]. For example, a posterior probability allows decisions that can use an utility model [1]. Posterior probabilities are also required when a classifier is making a small part of an overall decision, and the classification outputs must be combined for the overall decisions [3–6]. However, for a sample x, the output of the powerful support vector machine (SVM) [2,7,8] is a score $f(x)$, instead of a probability. These scores can be used to rank the samples in the test set from the most probable member to the least probable member of the positive class Y, i.e., for two samples x_i and x_j, if $f(x_i) < f(x_j)$, than $P(Y|x_i) < P(Y|x_j)$. In order to give the probability outputs for SVM, Wahba [9] and Platt [3] proposed SVM probability output models based on the output scores $f(x)$. Their models promote the applications of support vector machines [10–13] and have been extended to other binary SVM-type methods [14] and multi-class SVM methods [4,15].

In this paper, we are concerned about the probability estimates for twin support vector machine (TWSVM) [16]. TWSVM is a very efficient classifier [16,17] proposed recently. It is inspired by the non-parallel hyperplanes classifier [18], which aims at generating two non-parallel hyperplanes such that each hyperplane is closer to one of the two classes, and at the same time, is far from the other to some extent. Then TWSVM compares the distances between the sample and two non-parallel hyperplanes to determine which category the sample belongs to. A fundamental difference between TWSVM and SVM is that, TWSVM solves two small size quadratic programming problems (QPPs) instead of a large one in SVM. Comparing to SVM, TWSVM [16,17] is competitive in terms of performance whereas is around approximately four times faster. Experimental results in [16,17] showed the effectiveness of TWSVM over SVM on benchmark datasets. In addition, TWSVM is obviously much more effective than SVM at dealing with some special datasets, such as "Cross Planes" datasets. However, its output is not probability. Therefore, it is interesting to give the posterior probability outputs for TWSVM. But the current SVM probability models cannot be used directly because of the different mechanism of TWSVM and SVM.

In order to estimate the posterior probability, we propose a TWSVM probability output model, called PTWSVM. Our PTWSVM is obtained by the following three steps: Firstly, define a continuous output value by comparing the distances between the sample and two separating hyperplanes. This output value corresponds to the output score in SVM and can be used as the ranking score of the samples; the bigger the output value is, the more positive the sample is. Secondly, similar to [3], construct a sigmoid function with undetermined parameters where the input is the above continuous output value and the output is the posterior probability. Finally, calculate the above undetermined parameters in the sigmoid function by the maximum likelihood estimation. In addition, numerical results are also provided. Simulation results on the known probability distribution datasets validate that our PTWSVM works well. The experimental results on artificial and real world datasets show

* Corresponding author.
E-mail address: shaoyuanhai21@163.com (Y.-H. Shao).

0950-7051/$ - see front matter © 2012 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.knosys.2012.04.006

that our PTWSVM not only gives the proper probability output, but also improves the classification accuracy of TWSVM. This confirms that our PTWSVM is reasonable and effective.

The paper is organized as follows: In Section 2, we briefly introduce twin support vector machines. In Section 3, we propose our probability output model for TWSVM, including both linear and non-linear kernel cases. Computational comparisons on artificial and benchmark datasets are made in Section 4, and Section 5 gives some concluding remarks.

2. Twin support vector machine

Consider the binary classification problem of classifying m_1 training samples belonging to class +1 and m_2 training samples belonging to class -1 in the n-dimensional real space R^n, where $m_1 + m_2 = m$. Let matrix $A \in R^{m_1 \times n}$ represent the training samples of class +1 and matrix $B \in R^{m_2 \times n}$ represent the training samples of class -1.

Linear twin support vector machine (TWSVM) [16,17] seeks two non-parallel hyperplanes, a positive hyperplane $f_1(x) = 0$ and a negative hyperplane $f_2(x) = 0$

$$f_1(x) = w_1^\top x + b_1 = 0 \text{ and } f_2(x) = w_2^\top x + b_2 = 0, \tag{1}$$

such that each hyperplane is closest to training samples of one class and far from the training samples of the other class. The above two hyperplanes are obtained by solving the following two quadratic programming problems

$$\begin{aligned} \min_{w_1,b_1,\xi_2} \quad & \frac{1}{2}\|Aw_1 + e_1b_1\|^2 + c_1e_2^\top \xi_2, \\ \text{s.t.} \quad & -(Bw_1 + e_2b_1) + \xi_2 \geqslant e_2, \xi_2 \geqslant 0, \end{aligned} \tag{2}$$

and

$$\begin{aligned} \min_{w_2,b_2,\xi_1} \quad & \frac{1}{2}\|Bw_2 + e_2b_2\|^2 + c_2e_1^\top \xi_1, \\ \text{s.t.} \quad & (Aw_2 + e_1b_2) + \xi_1 \geqslant e_1, \xi_1 \geqslant 0, \end{aligned} \tag{3}$$

where $c_1 > 0$ and $c_2 > 0$ are parameters, e_1 and e_2 are vectors of ones of appropriate dimensions, and $\|\cdot\|$ denotes the L_2 norm. Their geometric meaning is clear. For example, for (2), its objective function makes class +1 proximity to the hyperplane $w_1^\top x + b_1 = 0$, while the constraints make class -1 bounded in the hyperplane $w_1^\top x + b_1 = -1$. For the above two QPPs, if samples of both the class are approximately equal to $m/2$, the complexity of solving these two QPPs of TWSVM is $O(2 \times (m/2)^3)$. Thus compare with the standard SVM (Its computational complexity is $O(m^3)$), the ratio of run times between the standard SVM and TWSVM is $m^3/(2 \times (m/2)^3)$. This makes TWSVM approximately four times faster [16] than standard SVM with the assumption that samples of both the classes are equal.

It has been shown that when both $P^\top P$ and $Q^\top Q$ are positive definite, the Wolfe duals of (2) and (3) have been shown to be

$$\begin{aligned} \max_{\alpha} \quad & e_2^\top \alpha - \frac{1}{2}\alpha^\top P(Q^\top Q)^{-1}P^\top \alpha \\ \text{s.t.} \quad & 0 \leqslant \alpha \leqslant c_1e_2, \end{aligned} \tag{4}$$

and

$$\begin{aligned} \max_{\gamma} \quad & e_1^\top \gamma - \frac{1}{2}\gamma^\top Q(P^\top P)^{-1}Q^\top \gamma \\ \text{s.t.} \quad & 0 \leqslant \gamma \leqslant c_2e_1, \end{aligned} \tag{5}$$

respectively, where $P = [B\ e_2]$, $Q = [A\ e_1]$, $\alpha \in R^{m_2}$ and $\gamma \in R^{m_1}$ are Lagrangian multipliers. In order to deal with the case when $P^\top P$ or $Q^\top Q$ is singular and avoid the possible ill-conditioning of $P^\top P$ and $Q^\top Q$, TWSVM introduces a term εI ($\varepsilon > 0$), where I is an identity matrix of appropriate dimensions. Therefore, the non-parallel hyperplanes (1) can be obtained from the solutions α and γ of (4) and (5) by

$$z_1 = -(Q^\top Q + \varepsilon I)^{-1}P^\top \alpha, z_2 = (P^\top P + \varepsilon I)^{-1}Q^\top \gamma, \tag{6}$$

where $z_k = [w_k^\top\ b_k]^\top$, $(k = 1,2)$.

Once the solutions (w_1,b_1) and (w_2,b_2) of the above problems are obtained, a new sample $x \in R^n$ is assigned to class i ($i = +1,-1$), depending on which of the two hyperplanes given by (1) it lies closest to, i.e.,

$$\text{Class } i = \arg \min_{k=1,2} \frac{|w_k^\top x + b_k|}{\|w_k\|}, \tag{7}$$

where $|\cdot|$ is the absolute value.

Linear TWSVM has also been extended in [16] to handle nonlinear discrimination by considering following two kernel generated surfaces

$$f_1(x) = K(x^\top, C^\top)u_1 + b_1 = 0 \text{ and } f_2(x) = K(x^\top, C^\top)u_2 + b_2 = 0, \tag{8}$$

where $C^\top = [A\ B]^\top$ and K is an appropriately chosen kernel. The corresponding optimization problems are

$$\begin{aligned} \min_{u_1,b_1,\xi_2} \quad & \frac{1}{2}\|K(A,C^\top)u_1 + e_1b_1\|^2 + c_1e_2^\top \xi_2, \\ \text{s.t.} \quad & -(K(B,C^\top)u_1 + e_2b_1) + \xi_2 \geqslant e_2, \\ & \xi_2 \geqslant 0, \end{aligned} \tag{9}$$

and

$$\begin{aligned} \min_{u_2,b_2,\xi_1} \quad & \frac{1}{2}\|K(B,C^\top)u_2 + e_2b_2\|^2 + c_2e_1^\top \xi_1, \\ \text{s.t.} \quad & (K(A,C^\top)u_2 + e_1b_2) + \xi_1 \geqslant e_1, \\ & \xi_1 \geqslant 0, \end{aligned} \tag{10}$$

where $c_1 > 0$ and $c_2 > 0$ are parameters.

Once the solutions (u_1,b_1) and (u_2,b_2) of the above problems are obtained, a new sample $x \in R^n$ is assigned to class i ($i = +1,-1$) in a similar manner to the linear case, i.e.,

$$\text{Class } i = \arg \min_{k=1,2} \frac{|K(x,C^\top)u_k + b_k|}{\sqrt{u_k^\top K(C,C^\top)u_k}}. \tag{11}$$

As stated above, compare with SVM, TWSVM has the superiorly on dealing with the "Cross Planes" type data [16,17]. As an example, we give a two-dimensional "Cross Planes" dataset in Fig. 1, the "Cross Planes" dataset is generated by perturbing points lying on two intersecting lines. Fig. 1 shows the dataset and the linear classifiers obtained by TWSVM and SVM. We get the quite different accuracy of these two classifiers (Accuracies are 99.01%(TWSVM) and 88.89%(SVM) by using 10-fold cross-validation). From Fig. 1, it is easy to see that the result of TWSVM is more reasonable than that of SVM. For strong generalization ability, TWSVMs have been preliminarily applied to some real-world problems such as text categorization and bioinformatics [19–21], and the methods of constructing the non-parallel hyperplanes have been studied extensively [22–27].

3. TWSVM probability model

In this section, we establish our TWSVM probability model. Firstly, we consider linear TWSVM, where a positive hyperplane $w_1^\top x + b_1 = 0$ and a negative hyperplane $w_2^\top x + b_2 = 0$ are constructed and new sample will be assigned to class either +1 or -1 depending on its proximity to the two hyperplanes. More precisely, the sample x is assigned to class +1 (or -1) if

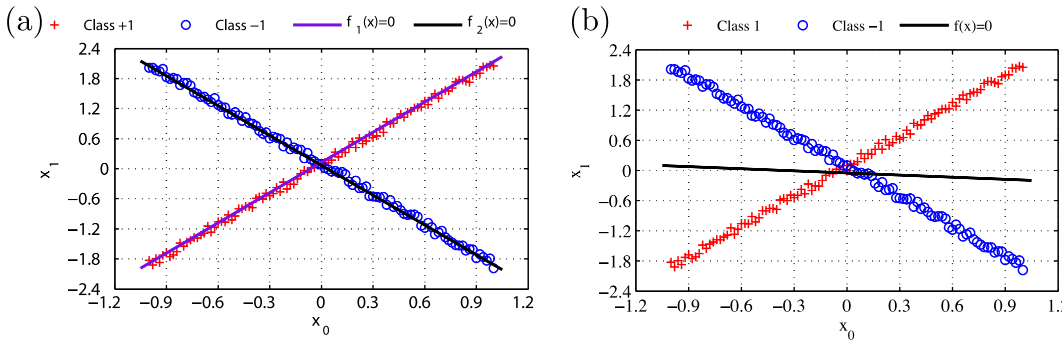

Fig. 1. TWSVM (a) and SVM (b) for "Cross Planes" dataset.

$\frac{|w_1^\top x+b_1|}{|w_2^\top x+b_2|}\cdot\frac{\|w_2\|}{\|w_1\|}<1$ (or $\frac{|w_1^\top x+b_1|}{|w_2^\top x+b_2|}\cdot\frac{\|w_2\|}{\|w_1\|}>1$). Now we hope to know what kind of samples are more likely belongs to class +1 (or class −1), or to what extent we can believe that the sample belongs to class +1 (or class −1). Intuitively speaking, the probability of a sample x belonging to a certain class depends on its location relative to the separating hyperplanes

$$\frac{w_1^\top x+b_1}{\|w_1\|}+\frac{w_2^\top x+b_2}{\|w_2\|}=0, \tag{12}$$

and

$$\frac{w_1^\top x+b_1}{\|w_1\|}-\frac{w_2^\top x+b_2}{\|w_2\|}=0. \tag{13}$$

A careful observation leads to consider the two relevant quantities $d(x)$ and $\frac{d(x)}{D(x)}$ which are defined by

$$d(x)=\min\{D_+(x),D_-(x)\}, \tag{14}$$

and

$$D(x)=\max\{D_+(x),D_-(x)\}, \tag{15}$$

where $D_+(x)$ and $D_-(x)$ are the distances from the sample x to the separating hyperplanes (12) and (13)

$$D_+(x)=\frac{\left|\frac{w_1^\top x+b_1}{\|w_1\|}+\frac{w_2^\top x+b_2}{\|w_2\|}\right|}{\left\|\frac{w_1}{\|w_1\|}+\frac{w_2}{\|w_2\|}\right\|}=\frac{\left|\frac{w_1^\top x+b_1}{\|w_1\|}+\frac{w_2^\top x+b_2}{\|w_2\|}\right|}{\sqrt{2+\frac{2(w_1^\top w_2)}{\|w_1\|\|w_2\|}}}, \tag{16}$$

$$D_-(x)=\frac{\left|\frac{w_1^\top x+b_1}{\|w_1\|}-\frac{w_2^\top x+b_2}{\|w_2\|}\right|}{\left\|\frac{w_1}{\|w_1\|}-\frac{w_2}{\|w_2\|}\right\|}=\frac{\left|\frac{w_1^\top x+b_1}{\|w_1\|}-\frac{w_2^\top x+b_2}{\|w_2\|}\right|}{\sqrt{2-\frac{2(w_1^\top w_2)}{\|w_1\|\|w_2\|}}}, \tag{17}$$

see Fig. 2. It can be seen that, for a sample x which is assigned to class +1, the quantities $d(x)$ and $\frac{d(x)}{D(x)}$ are factors influencing the degree belonging to the positive class because it should be more likely belongs to class +1 when either $d(x)$ or $\frac{d(x)}{D(x)}$ becomes larger. So it is natural to imagine the probability of the sample x belonging to the positive class depends on

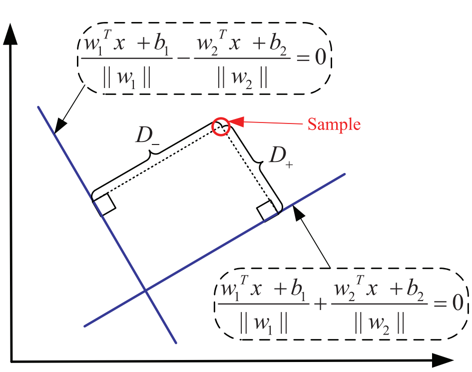

Fig. 2. An example indicating the meaning of the $D_+(x)$ and $D_-(x)$ in two-dimensional plane. Solid line stands for the separating hyperplanes (bounds) $\frac{w_1^\top x+b_1}{\|w_1\|}+\frac{w_2^\top x+b_2}{\|w_2\|}=0$ and $\frac{w_1^\top x+b_1}{\|w_1\|}-\frac{w_2^\top x+b_2}{\|w_2\|}=0$.

$$f(x)=\begin{cases} d(x)\left(\frac{d(x)}{D(x)}\right)^\alpha & \frac{|w_1^\top x+b_1|}{|w_2^\top x+b_2|}\cdot\frac{\|w_2\|}{\|w_1\|}<1,\\ 0 & \frac{|w_1^\top x+b_1|}{|w_2^\top x+b_2|}\cdot\frac{\|w_2\|}{\|w_1\|}=1,\\ -d(x)\left(\frac{d(x)}{D(x)}\right)^\alpha & \frac{|w_1^\top x+b_1|}{|w_2^\top x+b_2|}\cdot\frac{\|w_2\|}{\|w_1\|}>1, \end{cases} \tag{18}$$

where $\alpha > 0$ is a weight parameter. The bigger the value $f(x)$ is, the more positive the x is, and vice versa. Obviously the range of the values of the function (18) is $(-\infty,+\infty)$.

Now, we take a toy example (Example 1) to show that $f(x)$ is a suitable continuous ranking output for TWSVM. Fig. 3a shows a two-dimensional uniform distribution dataset, where "+" represents class +1 and "∘" represents class −1. Fig. 3a also shows the positive and negative hyperplanes learned by linear TWSVM (Accuracy is 100%(TWSVM) by using 10-fold cross-validation). Fig. 3b shows the values $f(x)$ calculated by (18), where $\alpha = 1$. From Fig. 3a, we obtain that TWSVM is the totally correct on this example, the contour of $f(x)$ reflects the distance between the sample and the bounds. In addition, from Fig. 3b, for positive and negative samples, the farther from the separating hyperplanes, the greater output value $|f(x)|$ is. This is to say that $f(x)$ is a suitable continuous output for TWSVM.

Platt et al. [3,4] has shown empirically that the sigmoid method yields probability estimates that are at least as accurate as ones obtained by training a SVM specifically for producing accurate class membership probability estimates, while being faster. Similar to the continuous output in SVM, the values range of $f(x)$ in our model also from negative infinity to positive infinity. So, instead of directly estimating the a posteriori probability in TWSVM, following [3,4], we construct

$$P(y=+1|f(x))=\frac{1}{1+e^{af(x)+b}}, \tag{19}$$

$$P(y=-1|f(x))=1-P(y=+1|f(x)), \tag{20}$$

where a and b are parameters. Following the maximum likelihood estimation [3], the parameters a and b in (19) and (20) as well as the weight parameter alpha in (18) can be obtained by using many regularization methods. For simplicity, setting $\alpha = 1$, the parameters a and b are found by minimizing the following function

$$\begin{aligned} \min_{z=(a,b)}\quad & F(z)=-\sum_{i=1}^{m}(t_i\log p_i+(1-t_i)\log(1-p_i)),\\ \text{s.t.}\quad & p_i=P(y=+1|f(x)),\\ & t_i=\begin{cases}\frac{m_1+1}{m_1+2} & if\quad y_i=+1\\ \frac{1}{m_2+2} & if\quad y_i=-1\end{cases}\quad i=1,\ldots,m, \end{aligned} \tag{21}$$

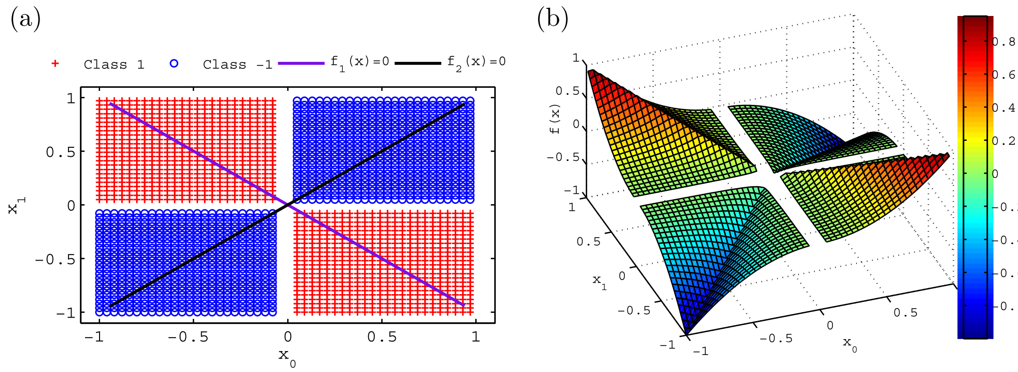

Fig. 3. (a): TWSVM classify on Example 1; (b): $f(x)$ calculated by (18) on Example 1.

where m, m_1 and m_2 are the number of the training samples, positive training samples and negative training samples, respectively. The advantage of using this criterion to adjust the sigmoid parameters are that the Hessian $H(a,b)=\bigtriangledown^2 F(a,b)$ is positive definitive, so the problem can be solved using Newton methods without any risks of finding local minima. Here, we use the technique of [28,29] to solve (21). It should be point out that when the positive samples are equals to negative samples, put the positive samples as negative ones in (21), the results are the same.

The above results can be easily extended to non-linear TWSVM by considering the kernel-generated surfaces (8) instead of the hyperplanes (1). Similar to the linear case, we also introduce $d(x)$ and $\frac{d(x)}{D(x)}$, where $d(x)=\min\{D_+(x),D_-(x)\}$ and $D(x)=\max\{D_+(x),D_-(x)\}$, $D_+(x)$ and $D_-(x)$ are the distances from a sample to the separating surfaces

$$\frac{u_1^\top K(x,C^\top)+b_1}{\sqrt{u_1^\top K(C,C^\top)u_1^\top}}+\frac{u_2^\top K(x,C^\top)+b_2}{\sqrt{u_2^\top K(C,C^\top)u_2^\top}}=0, \tag{22}$$

$$\frac{u_1^\top K(x,C^\top)+b_1}{\sqrt{u_1^\top K(C,C^\top)u_1^\top}}-\frac{u_2^\top K(x,C^\top)+b_2}{\sqrt{u_2^\top K(C,C^\top)u_2^\top}}=0, \tag{23}$$

$D_+(x)$ and $D_-(x)$ can be expressed as

$$D_+(x)=\frac{\left|\frac{u_1^\top K(x,C^\top)+b_1}{\sqrt{u_1^\top K(C,C^\top)u_1^\top}}+\frac{u_2^\top K(x,C^\top)+b_2}{\sqrt{u_2^\top K(C,C^\top)u_2^\top}}\right|}{\sqrt{2+\frac{2u_1^\top K(C,C^\top)u_2^\top}{u_1^\top K(C,C^\top)u_1^\top u_2^\top K(C,C^\top)u_2^\top}}}, \tag{24}$$

$$D_-(x)=\frac{\left|\frac{u_1^\top K(x,C^\top)+b_1}{\sqrt{u_1^\top K(C,C^\top)u_1^\top}}-\frac{u_2^\top K(x,C^\top)+b_2}{\sqrt{u_2^\top K(C,C^\top)u_2^\top}}\right|}{\sqrt{2-\frac{2u_1^\top K(C,C^\top)u_2^\top}{u_1^\top K(C,C^\top)u_1^\top u_2^\top K(C,C^\top)u_2^\top}}}. \tag{25}$$

Then, the degree of the sample x belonging to the positive (or negative) class depends on

$$f(x)=\begin{cases} d(x)\left(\frac{d(x)}{D(x)}\right)^\alpha & \frac{|u_1^\top K(x,C^\top)+b_1|}{|u_2^\top K(x,C^\top)+b_2|}\frac{\sqrt{u_2^\top K(C,C^\top)u_2^\top}}{\sqrt{u_1^\top K(C,C^\top)u_1^\top}}<1;\\ 0 & \frac{|u_1^\top K(x,C^\top)+b_1|}{|u_2^\top K(x,C^\top)+b_2|}\frac{\sqrt{u_2^\top K(C,C^\top)u_2^\top}}{\sqrt{u_1^\top K(C,C^\top)u_1^\top}}=1;\\ -d(x)\left(\frac{d(x)}{D(x)}\right)^\alpha & \frac{|u_1^\top K(x,C^\top)+b_1|}{|u_2^\top K(x,C^\top)+b_2|}\frac{\sqrt{u_2^\top K(C,C^\top)u_2^\top}}{\sqrt{u_1^\top K(C,C^\top)u_1^\top}}>1. \end{cases} \tag{26}$$

Similar to the linear case, the continuous output $f(x)$ in (26) also can be ranked, and the larger the value $f(x)$ is, the more positive the x is, and vice versa. The class-conditional densities estimation in non-linear TWSVM is the same with the linear one.

4. Experimental results

To demonstrate the performance of our probability output model, we conducted experiments on an artificial dataset and seven UCI benchmark data sets [30]. In our experiments, all methods were implemented in Malab 7.0 [31] environment on a PC with an Intel P4 processor (2.9 GHz) with 1 GB RAM. The dual QPPs arising in TWSVM were solved by the optimization toolbox QP in Matlab [31]. The "Accuracy" used to evaluate methods is defined as follows: $Accuracy=(TP+TN)/(TP+FP+TN+FN)$, where TP, TN, FP and FN are the number of true positive, true negative, false positive and false negative, respectively. As for the problem of selecting hyper-parameters, we employ standard 10-fold cross-validation technique [32]. Furthermore, the parameters for the all methods are selected from the set $\{2^{-8},\ldots,2^{8}\}$.

4.1. Toy example

In this section, we give a toy example (Example 2) to show the class-conditional densities estimation by our PTWSVM is an apparent exponential. We generate 100 positive and negative samples in 1-dimension with Gaussian distribution $x=\frac{1}{\sigma\sqrt{2\pi}}e^{\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}$, respectively. Here $\mu=-1$ and $\sigma=2$ for positive samples, $\mu=1$ and $\sigma=2$ for negative samples. Then Bayes' rule is used to calculate the probability of the samples belongs to positive (and negative) class as

$$P(y=1|f)=\frac{p(f|y=1)p(y=1)}{\sum_{i=-1,1}p(f|y=i)p(y=i)}, \tag{27}$$

where $p(y=i)$ is prior probability that can be computed from the training samples. We also use linear TWSVM to classify this dataset and use PTWSVM to calculate the probability output. In TWSVM, the best parameters are $c_1=0.1$ and $c_2=1$, and Accuracy is 84.50%. In our PTWSVM, $f(x_i)$ is calculated with $a=-5.3283$ and $b=0.5325$. Fig. 4a shows the output $f(x_i)$ and probability output calculated by PTWSVM. From Fig. 4a, we obtain that the PTWSVM obeys sigmoid distribution well. Fig. 4b shows the probability output calculated by Bayes' rule and PTWSVM, respectively. From Fig. 4, we can see that our PTWSVM simulates the posterior probability similar to the Bayes' rule on Example 2.

4.2. UCI datasets

In this section, we further experimented with benchmark datasets to show the class-conditional densities estimation by our PTWSVM is an apparent exponential. Table 1 lists the details of seven UCI benchmark datasets [30]. At first, we test our PTWSVM on Twonorm and Adult datasets. In our experiments, Twonorm and

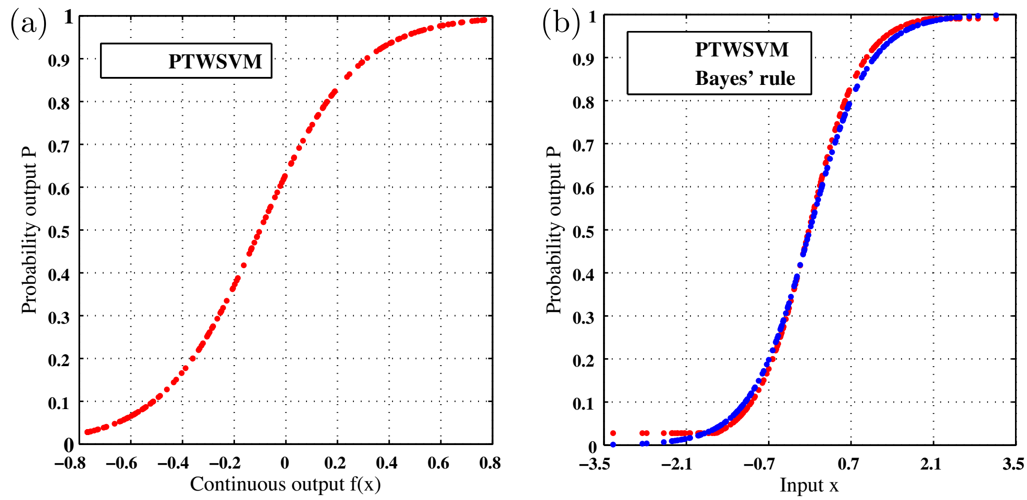

Fig. 4. The probability output on the Example 2. In (a), the *x*-coordinate is the $f(x_i)$ calculate by PTWSVM, and the *y*-coordinate is the probability, and red points is the best-fit sigmoid to the posterior, using the algorithm described in this paper. In (b), the *x*-coordinate is the input samples, and the *y*-coordinate is the probability, blue circle is the posterior probability computed by (27), and red points is the best-fit sigmoid to the posterior, using the algorithm described in this paper.

Table 1
Descriptions of benchmark datasets used in experiments.

Datasets	Training samples	Test samples	Positive %	Negative %	Feature
Titanic	150	2051	58.33	41.67	3
Diabetics	468	300	34.90	65.10	8
German	700	300	30	70	20
Ringnorm	400	7000	49.51	50.49	20
Twonorm	400	7000	50.04	49.96	20
Spambase	3221	1380	39.40	60.60	57
Adult	1605	16282	24.78	75.22	13

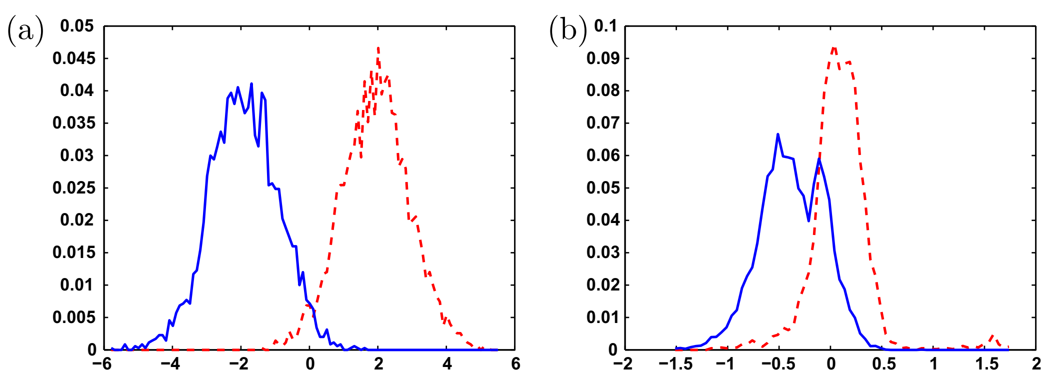

Fig. 5. The histograms for $P(f|y=\pm1)$ for a linear TWSVM trained on (a) "Twonrom" and (b) "Adult" data set. The solid line is $P(f|y=-1)$, while the dashed line is $P=(f|y=+1)$.

Adult datasets are divided into training set with 400 samples and 1605 samples, and test set with 7000 samples and 16,282 samples, respectively. Fig. 5a and b shows the plot of the class-conditional densities $P(f|y=\pm1)$ of Twonorm and Adults datasets, which trained by linear TWSVM. The plot shows histograms of the densities (with bins 0.1 wide on Twonorm dataset and with bins 0.05 wide on Adult dataset). From Fig. 5, we can see that the densities are close to Gaussian on Twonorm dataset, while the densities are far away from Gaussian on Adult dataset. There are discontinuities in the derivatives of both densities at $f=0$. This is because that the cost function of TWSVM has discontinuities at the bounds.

For these two different type densities, we calculate the probability output by PTWSVM. Fig. 6 shows reliability diagrams for sigmoid functions to Two-Norm and Adult datasets calculated by PTWSVM, where the solid line is the best-fit sigmoid to the posterior. From Fig. 6, it is easy to see that the sigmoid fit by PTWSVM is similar to Example 2 both for Two-Norm and Adult datasets. That is to say, our PTWSVM not only can fit well on Gaussian densities, but also on other type densities. From the above results, we can appreciated that, given the set of samples, our PTWSVM is suitable election.

In addition, we compare our PTWSVM to a plain TWSVM for misclassification rate. Specifically, we set the optimal threshold $P(y=+1|f)=0.5$ for the PTWSVM, while the optimal threshold for the TWSVM is $f=0$. Then, we check to see wether the threshold $f=0$ is optimal for TWSVM in the following experiments.

Table 2 summarizes the comparison of classification accuracy for our PTWSVM with that of TWSVM [16] for linear kernel on

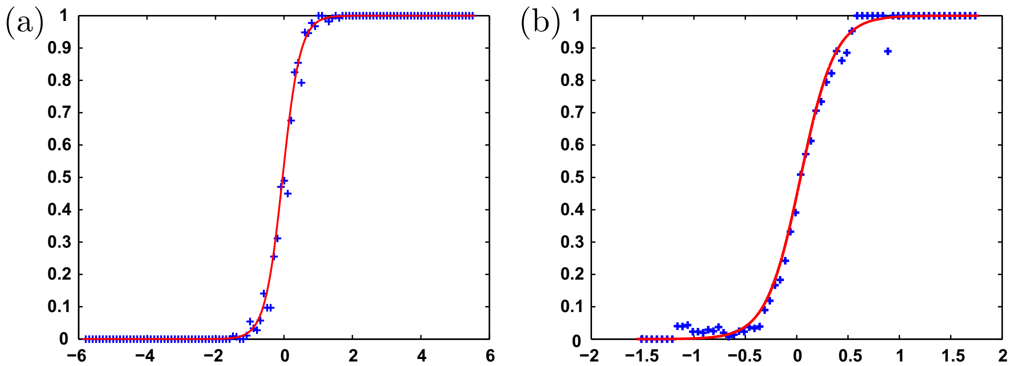

Fig. 6. The fit of the sigmoid to the data for linear TWSVM on (a) "Twonorm" and (b) "Adult" datasets. Each plus mark is the posterior probability computed for all samples falling into a bin of width 0.1 on "Twonorm" dataset and 0.05 on "Adult" dataset. The solid line is the best-fit sigmoid to the posterior, using the algorithm described in this paper.

Table 2
Ten-fold testing percentage accuracy of linear classifiers.

Dateset	TWSVM	PTWSVM	C_1/C_2	b/a
Titanic	77.74	**78.11**	1/0.5	1.0993
Diabetics	**81.00**	80.67	0.1/0.1	−0.0633
German	78.33	**80.00**	2/0.5	0.1813
Ring-norm	**75.93**	75.81	1/1	−0.2135
Two-norm	**97.37**	**97.37**	1/0.0625	0.0142
Spambase	90.36	**91.81**	3/1	−2.7381
Adult	83.10	**83.13**	0.25/4	−0.0044

Table 3
Ten-fold testing percentage accuracy of non-linear classifiers.

Dateset	TWSVM	PTWSVM	C_1/C_2	$p(\mu)$	B/A
Titanic	89.31	**90.10**	0.1/1	12	0.4645
Diabetics	**81.33**	80.67	0.1/0.1	1	−0.0679
German	**72.67**	**72.67**	0.05/0.25	3	3.0286
Ring-norm	98.17	**98.46**	0.1/0.2	4	0.0146
Two-norm	96.41	**97.27**	4/0.25	3	0.0129
Spambase	**99.93**	99.86	8/4	10	−0.2306
Adult	82.32	**83.37**	1/1	2	0.0150

seven benchmark datasets. Here, the best parameters $c_i\,(i=1,2)$ are searching from $\{2^{-8},\ldots,2^{8}\}$ in TWSVM. For each dataset, the parameters c_i of PTWSVM were set to be the same with TWSVM as shown in Table 2. Table 2 also shows the test accuracies and the parameters used in the TWSVM and PTWSVM, where the best accuracy is shown by bold figures. From Table 2, we can see that the generalization capability of PTWSVM (using $P(y=+1|f)=0.5$ as threshold) is better than TWSVM (using $f=0$ as threshold) on four datasets, while TWSVM is better than PTWSVM on three datasets considered with the same parameters. It indicates that threshold $f=0$ is not always an optimal in TWSVM.

Table 3 is concerned with kernel PTWSVM and TWSVM with polynomial kernel $K(x,x')=(xx'+1)^p$ on "Titanic", "Diabetics" and "German" datasets and with Gaussian kernel $K(x,x')=e^{-\mu\|x-x'\|^2}$ on the rest four datasets. The results in Table 3 are similar with that appeared in Table 2, and therefore confirm the above conclusion further.

By introducing PTWSVM, the receiver operating characteristic (ROC) curve of TWSVM can be drawn easily. ROC analysis provides the most comprehensive description of diagnostic accuracy available to date, because it estimates and reports all of the combinations of sensitivity and specificity that a diagnostic test is able to provide. As an example, we show the ROC curves for Twonorm dataset by using a linear kernel and Adult dataset by using a Gaussian kernel for TWSVM and SVM in Fig. 7, respectively. The True positive rate, False positive rate are defined as follows: *True positive rate* $=TP/(TP+FN)$, *False positive rate* $=1-TN/(TN+FP)$. From Fig. 7, we can see that the accuracy of TWSVM is higher than SVM on Twonorm dataset (Accuracies are 97.86%(TWSVM) and 96.96%(SVM) by using 10-fold cross-validation), and the AUC (the

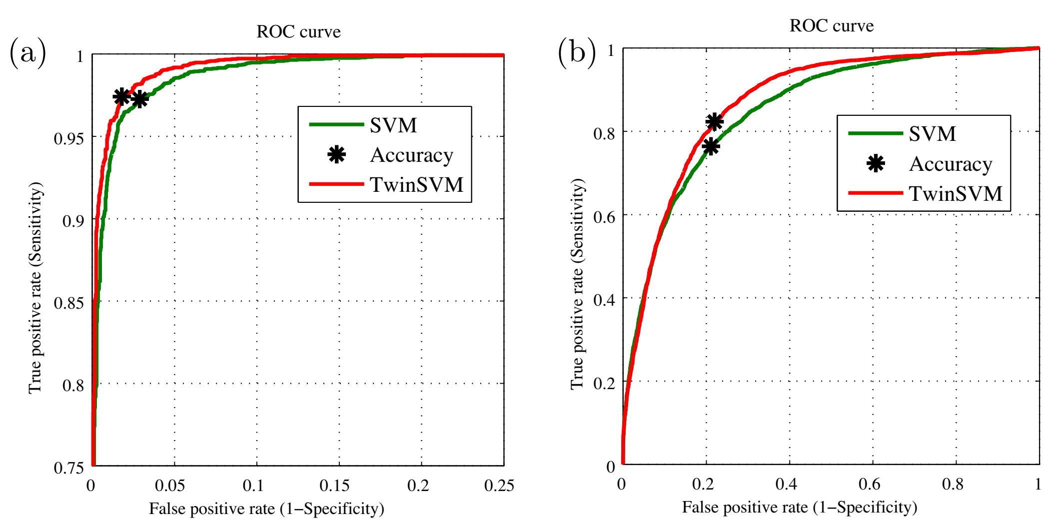

Fig. 7. TWSVM ROC curve for Twonorm dataset (a) and Adult dataset (b).

area under the ROC curve) of TWSVM is higher than SVM (AUC are 0.9975(TWSVM) and 0.9960 (SVM)). However, for Adult dataset, the accuracy of TWSVM is lower than SVM (Accuracies are 79.19%(TWSVM) and 81.94%(SVM) by using 10-fold cross-validation), but the AUC of TWSVM is higher than SVM (Accuracies are 0.87286(TWSVM) and 0.8547(SVM)). That is to say, by using ROC curves, we can analysis TWSVM in more detailed.

5. Conclusions

In this paper, we have proposed a novel TWSVM probability model, called PTWSVM, to estimate the posterior probability of the twin support vector machines (TWSVM). The key point is that, for a sample, define an output value corresponding to the output score in SVM. The preliminary experiments have demonstrated that our PTWSVM works well on both artificial and benchmark datasets, showing its rationality and effectiveness. Our PTWSVM MATLAB codes can be downloaded from http://www.cau.edu.cn/sci/index.php?do = xzshow& id = 177. The model also can be extended to some relevant methods, such as GEPSVM, LS-TSVM and NPPC [18,19,25,33–35] etc. A practical problem is the selection of the parameters in the model which should be addressed in the future. Furthermore, practical applications (such as [36–40]) are also interesting.

Acknowledgments

We thank the anonymous reviewers for their valuable suggestions. This work is supported by the National Natural Science Foundation of China (Nos. 10971223, 10926198, 11161045 and 11071252) and the Zhejiang Provincial Natural Science Foundation of China (No. Q12A010077, Y1100237).

References

[1] R.O. Duda, P.E. Hart, Pattern Classification and Scene Analysis, John Wiley & Sons, 1973.
[2] V.N. Vapnik, Statistical Learning Theory, Wiley, New York, 1998.
[3] J. Platt, Probabilistic outputs for support vector machines and comparison to regularized likelihood methods, in: Advances in Large Margin Classifiers, MIT Press, Cambridge, MA, 2000.
[4] B. Zadrozny, C. Elkan, Transforming classifier scores into accurate multiclass probability estimates, in: SIGKDD, 2002.
[5] B. Zadrozny, C. Elkan, Obtaining calibrated probability estimates from decision trees and naive bayesian classifiers, in: Proceedings of the Eighteenth International Conference on Machine Learning, Morgan Kaufmann Publishers, Inc., 2001, pp. 609–616.
[6] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, Heidelberg, 2005.
[7] C. Cortes, V.N. Vapnik, Support Vector Networks, Machine Learning 20 (1995) 273–297.
[8] N.Y. Deng, Y.J. Tian, C.H. Zhang, Support Vector Machines: Theory, Algorithms, and Extensions, CRC Press, 2012.
[9] G. Wahba, Support vector machines: reproducing Kernel Hilbert spaces and the randomized, in: B. Schölkopf, C.J.C. Burges, A.J. Smola (Eds.), Advances in Kernel Mehods-Support Vector Learning, Cambridge, MA, 1999, pp. 69–88.
[10] G. Obozinski, G. Lanckriet, C. Grant, M. Jordan, W. Noble, Consistent probabilistic output for protein function prediction, Genome Biology 9 (Suppl. 1) (2008) S6.
[11] K.Q. Shen, C.J. Ong, X.P. Li, E.P. Wilder-Smith, Feature selection via sensitivity analysis of SVM probabilistic outputs, Machine Learning 70 (1) (2008) 1–20.
[12] A. Ganapathiraju, J. Hamaker, J. Picone, Applications of support vector machines to speech recognition, IEEE Transactions on Signal Process 52 (8) (2004) 2348–2355.
[13] Q. Tao, G.W. Wu, F.Y. Wang, Posterior probability support vector machines for unbalanced data, IEEE Transactions on Neural Networks 16 (6) (2005) 1561–1573.
[14] G.T. Van, J. Suykens, G. Lanckriet, A. Lambrechts, M.B. De, J. Vandewalle, Multiclass LS-SVMs: moderated outputs and coding-decoding schemes, Neural Processing Letters 15 (2002) 45–58.
[15] T.F. Wu, C.J. Lin, Ruby C. Weng, Probability estimates for multi-class classification by pairwise coupling, Journal of Machine Learning Research 5 (2004) 975–1005.
[16] Jayadeva, R. Khemchandani, S. Chandra, Twin support vector machines for pattern classification, IEEE Transactions on Pattern Analysis and Machine Intelligence 29 (5) (2007) 905–910.
[17] Y.H. Shao, C.H. Zhang, X.B. Wang, N.Y. Deng, Improvements on twin support vector machines, IEEE Transactions on Neural Networks 22 (6) (2011) 962–968.
[18] O.L. Mangasarian, E.W. Wild, Multisurface proximal support vector classification via generalize deigenvalues, IEEE Transactions on Pattern Analysis and Machine Intelligence 28 (1) (2006) 69–74.
[19] M.A. Kumar, M. Gopal, Least squares twin support vector machines for pattern classification, Expert Systems with Applications 36 (4) (2009) 7535–7543.
[20] G.R. Naik, D.K. Kumar, Jayadeva, Twin SVM for gesture classification using the surface electromyogram, IEEE Transactions on Information Technology in Biomedicine 14 (2) (2010) 301–308.
[21] S. Ghorai, A. Mukherjee, S. Sengupta, P.K. Dutta, Cancer classification from gene expression data by NPPC ensemble, IEEE/ACM Transactions on Computational Biology and Bioinformatics 8 (3) (2011) 659–671.
[22] M.A. Kumar, M. Gopal, Application of smoothing technique on twin support vector machines, Pattern Recognition Letters 29 (13) (2008) 1842–1848.
[23] Y.H. Shao, N.Y. Deng, Z.M. Yang, Least squares recursive projection twin support vector machine for classification, Pattern Recognition 45 (6) (2012) 2299–2307.
[24] Y.H. Shao, N.Y. Deng, A coordinate descent margin based-twin support vector machine for classification, Neural Networks 25 (2012) 114–121.
[25] S. Ghorai, A. Mukherjee, P.K. Dutta, Nonparallel plane proximal classifier, Signal Processing 89 (4) (2009) 510–522.
[26] X. Peng, A ν-twin support vector machine (ν-TSVM) classifier and its geometric algorithms, Information Sciences 180 (2010) 3863–3875.
[27] S. Ghorai, S.J. Hossain, A. Mukherjee, P.K. Dutta, Newton's method for nonparallel plane proximal classifier, Signal Processing 90 (2010) 93–104.
[28] H.T. Lin, C.J. Lin, R.C. Weng, A note on Platt's probabilistic outputs for support vector machines, Machine Learning 68 (3) (2007) 267–276.
[29] T.K. Huang, R.C. Weng, C.J. Lin, Generalized Bradley-terry models and multi-class probability estimates, Journal of Machine Learning Research 7 (2006) 85–115.
[30] C.L. Blake, C.J. Merz, UCI Repository for Machine Learning Databases, Dept. of Information and Computer Sciences, Univ. of California, Irvine, 1998. <http://www.ics.uci.edu/mlearn/MLRepository.html>.
[31] MATLAB, 2007. <http://www.mathworks.com>.
[32] R.O. Duda, P.E. Hart, D.G. Stork, Pattern Classification, second ed., John Wiley and Sons, 2001.
[33] Y.H. Shao, Z.X. Yang, X.B. Wang, N.Y. Deng, Multiple instances twin support vector machines, in: The Ninth International Symposium on Operations Research and Its Applications, China, September 2010, pp. 433–442.
[34] Y.H. Shao, N.Y. Deng, A novel margin based twin support vector machine with unity norm hyperplanes, Neural Computing and Applications (2012), http://dx.doi.org/10.1007/s00521-012-0894-5.
[35] Q.L. Ye, C.X. Zhao, N. Ye, X.B. Chen, Localized twin SVM via convex minimization, Neurocomputing 74 (4) (2011) 580–587.
[36] C.F. Tsai, K.C. Cheng, Simple instance selection for bankruptcy prediction, Knowledge-Based Systems 27 (2012) 333–342.
[37] H.L. Chen, B. Yang, G. Wang, J. Liu, X. Xu, S.J. Wang, D.Y. Liu, A novel bankruptcy prediction model based on an adaptive fuzzy k-nearest neighbor method, Knowledge-Based Systems 24 (2011) 1348–1359.
[38] Y. Pan, H.X. Luo, Y. Tang, C.Q. Huang, Learning to rank with document ranks and scores, Knowledge-Based Systems 24 (2011) 478–483.
[39] D. Olszewski, A probabilistic approach to fraud detection in telecommunications, Knowledge-Based Systems 26 (2012) 246–258.
[40] Z.Q. Qi, Y.T. Xu, L.S. Wang, Y. Song, Online multiple instance boosting for object detection, Neurocomputing 74 (10) (2011) 1769–1775.